

Synthetic Data – A Privacy Mirage

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Abstract—Synthetic datasets drawn from generative models have been advertised as a silver-bullet solution to privacy-preserving data publishing. In this work, we show through an extensive privacy evaluation that such claims do not match reality. First, synthetic data does not prevent attribute inference. Any data characteristics preserved by a generative model for the purpose of data analysis, can simultaneously be used by an adversary to reconstruct sensitive information about individuals. Second, synthetic data does not protect against linkage attacks. We demonstrate that high-dimensional synthetic datasets preserve much more information about the raw data than the features in the model’s lower-dimensional approximation. This rich information can be exploited by an adversary even when models are trained under differential privacy. Moreover, we observe that some target records receive substantially less protection than others and that the more complex the generative model, the more difficult it is to predict which targets will remain vulnerable to inference attacks. Finally, we show why generative models are unlikely to ever become an appropriate solution to the problem of privacy-preserving data publishing.

I. INTRODUCTION

The rise of data-driven decision making as the prevailing approach to advance science, industrial production, and governance generates a need to share and publish data [1]–[3]. At the same time, concerns about the implications that open data sharing has for individuals and communities call for data publishing approaches that respect and preserve privacy rights.

Yet, how to share high-dimensional data in a privacy-preserving manner remains an unsolved problem. Attempts to anonymise data have failed across the board [4]–[10]. Technologies based on the differential privacy (DP) model [11], [12] focus mostly on (non-interactive) data analysis techniques [13]–[16], and often incur a significant utility loss [17]–[21]. In this landscape, practitioners [22]–[26] and researchers [27]–[29] see in synthetic data [30] a promising solution to avoid privacy issues. The majority of these efforts focus on synthetic data drawn from generative models [29], [31]–[34]. The promise of synthetic data is that it preserves raw data characteristics that permit analysts to draw the same statistical conclusions as when using the raw data but prevent the inference of sensitive information [24], [27], [28], [30].

Over the past few years, this promise became an appealing proposition to practitioners eagerly looking for solutions to the microdata release problem. From big public sector organisations such as NHS England [26] and the National Institute of Standards and Technology [22] to a growing number of companies [35]–[39], synthetic data has been hailed as the holy grail of privacy-preserving data publishing as it “contains no personal data, making it suitable to release as open data” [24] and “enable[s] the protection of personally

identifiable information (PII) while maintaining a dataset’s utility for analysis” [22].

Claims about the privacy benefits of synthetic data, however, have not been supported by rigorous assessments of its privacy properties. Both, institutions [22], [23], [25], and companies [35]–[39], assume that data synthesis, or a combination of data generating algorithms with DP techniques, are sufficient to provide strong privacy protection. We find three existing approaches to backup these claims:

I) Works that rely on similarity tests to assert that no one record in the synthetic data is “too close” to the original data [28], [40], [41].

II) Works that rely on formal privacy guarantees to assess the privacy gain of synthetic data releases [42], [43].

III) More recent works that introduce inference attacks on generative models that aim to extract sensitive information from a trained model [44]–[47].

In this paper, we demonstrate that *none of these approaches provides an appropriate framework to assess the privacy risks of synthetic data publishing*. First, privacy definitions based on data similarity do not capture privacy leakage through inference attacks and hence underestimate privacy risks. Furthermore, these privacy definitions conflate the boundaries between valid inference and privacy leakage: the utility of synthetic data can equally be measured as the distance between real and synthetic records. Second, formal privacy guarantees for a model’s lower-dimensional data representation do not necessarily translate into privacy guarantees against inference attacks on the model’s output. Besides, formal notions are hard to link to legal data protection requirements [48], [49]. Third, model extraction attacks assume that the adversary has oracle access to the model and can query it several times to perform her inference. This process, *does not* correspond to the data sharing scenario where typically the data holder will train the model locally, release a synthetic dataset, and keep the model private. Furthermore, most of these studies focus on non-parametric models trained on image datasets [45]–[47], which are rarely the subject of data sharing use cases. Given the current understanding of these models [50]–[52], it is doubtful whether their findings will generalise to other domains.

In this paper, we propose a novel evaluation framework that addresses the shortcomings of previous evaluation approaches. It inherently models the risk of inference attacks, it can be applied to assess any type of generative model, with and without formal privacy guarantees, and measures the risk of releasing a synthetic dataset rather than a trained model. We *quantitatively* assess the privacy properties of three generative

models and two differentially private variants on three realistic datasets. Our results challenge previous claims about the privacy benefits of synthetic data publishing and demonstrate that generative models often do not provide robust protection against privacy attacks on the generated datasets. In summary, our paper makes the following contributions:

We demonstrate that synthetic data does not provide robust, trade-off free, protection against attribute inference and linkage attacks. Our evaluation shows that publishing a synthetic dataset in place of the raw data provides little gain with respect to an adversary that aims to infer a sensitive attribute value from a partially known target record. Any gain in privacy with respect to this adversary comes at a necessary cost in utility. Synthetic data does not prevent linkage attacks. An adversary can infer with high confidence the presence of a target record in the training data of the generative model. This is because high-dimensional synthetic datasets carry much more information about the raw data than the features explicitly captured by the model’s lower-dimensional approximation, which in turn makes it impossible to predict the level of protection synthetic data will provide.

We demonstrate that differentially private model training does not increase privacy gain wrt linkage attacks. Differentially private training protects against ML-based linkage attacks that target features in the model’s lower-dimensional data representation. We demonstrate, however, that synthetic datasets generated by these models preserve enough information about the model’s training set for an adversary to link records to the training dataset.

We provide an open-source privacy evaluation framework. We introduce the first evaluation framework to quantify the privacy gain of publishing synthetic data drawn from a generative model in-place of (a sanitised version of) the original data that captures the risk of inference attacks and can be applied to *any type of generative model* training algorithm. We implement our evaluation framework as a modular open-source Python library. We instantiate it with two new privacy attacks that capture linkage and attribute inference risks.

II. SYNTHETIC DATA AND GENERATIVE MODELS

In this section, we formalise the process of synthetic data generation. Table II in the Appendix summarises our notation.

Let $\mathcal{R}$ be a population of data records where each record $\mathbf{r} \in \mathcal{R}$ contains k attributes: $\mathbf{r} = (r_1, \dots, r_k)$. We denote the unknown joint probability distribution over the data domain of the population $\mathcal{R}$ as $\mathcal{D}_{\mathcal{R}} = \mathbf{P}[\mathbf{r}]$. We refer to $R \sim \mathcal{D}_{\mathcal{R}}^n$, a collection of n data records $\mathbf{r}_i$ sampled independently from $\mathcal{D}_{\mathcal{R}}$, as *raw dataset* which defines the data distribution $\mathcal{D}_R = \mathbf{P}[\mathbf{r}|R]$.

Synthetic data generation. The goal of a generative model is to learn a representation of the joint multivariate probability distribution of data records $\mathcal{D}_R$. The model training algorithm $\text{GM}(R)$ takes as input a raw dataset R , learns $\mathcal{D}_{f(R)}$, a lower-dimensional representation of the data distribution $\mathcal{D}_R$, and outputs a trained generative model $f(R)$. $f(R)$ is a stochastic

function that, without any input, generates synthetic records s , distributed according to $\mathcal{D}_{f(R)}$. We write $f(R) \sim \text{GM}(R)$ instead of $f(R) \leftarrow \text{GM}(R)$ to indicate that the training algorithm can be a stochastic and non-deterministic process. To generate a synthetic dataset S of m records, we call m times the generative model $f(R)$. We denote the process of sampling m independent records from the model as $S \sim \mathcal{D}_{f(R)}^m$.

Approximation by features. It is tempting to assume that the synthetic data “carries through all of the statistical properties, patterns and correlations in the [input] data” [53] and that the model $\mathcal{D}_{f(R)}$ provides a perfect representation of the data distribution $\mathcal{D}_R$. The trained model, however, only provides a *lower-dimensional approximation* of the true distribution of the raw data that retains *some* characteristics but can never preserve all of them. Which characteristics are captured, and how they are used to approximate $\mathcal{D}_R$, is determined by the generative model choice.

Statistical models, such as Bayesian networks [54], or Hidden Markov models [55], provide an explicit, parametric model of $\mathcal{D}_R$. The set of features statistical models extract from their training data is determined upfront.

Non-parametric models, such as generative adversarial networks (GANs) [56] or variational auto encoders (VAEs) [57], do not estimate a parametric likelihood function in order to generate new samples from $\mathcal{D}_{f(R)}$ [32]. Instead, which features of the input data are most relevant and how the model approximates $\mathcal{D}_R$ is implicitly determined during training [32].

The features a generative model uses to approximate $\mathcal{D}_R$ define what statistical patterns are extracted from the raw data R , and therefore which patterns will be replicated by a synthetic dataset $S \sim \mathcal{D}_{f(R)}$. Statistical models provide some control over what features will be preserved. It is not possible to *exclude*, however, that characteristics other than the pre-defined set of features are replicated in a synthetic dataset. Strong patterns that are not explicitly modelled by the model’s parametric likelihood function might still be preserved. For instance, synthetic data generated through independent sampling from a set of marginal frequency counts is expected to preserve independent marginals. However, if the raw data contains strong correlations between attributes these correlations are likely to be replicated in the synthetic data even under independent sampling. Non-parametric models make it even harder to predict which features of a synthetic dataset S mimic original patterns in the raw data and which features are mere sampling noise. As our evaluation in Section VIII-A reveals, not being able to predict the full set of data features synthetic data preserves, has negative implications for the privacy protection it provides to the raw data.

III. EVALUATING THE PRIVACY GAIN OF SYNTHETIC DATA

Similar to claims about the unlimited utility of synthetic data [58], wide-ranging promises have been made about the “absolute privacy” [26] synthetic data provides. The most prominent argument to support claims about its privacy benefits is based on the idea that synthetic data is “artificial data” [22], [53] and no direct link exists between real and

synthetic records. Thus, many argue [24], [26], [28], [41], [53], synthetic data, by design, protects against traditional attacks on microdata releases such as linkage [59], [60] or attribute disclosure [60], [61]. More recently, the use of differentially private model training techniques has led many to conclude that synthetic data generated by differentially private models does not “contain any information that can be traced back to specific individuals in the original data” [22].

Both of these privacy arguments, however, seem to ignore the fact that *synthetic data that is useful for analysis must always leak some information* about the raw data and that a malicious adversary can leverage the preserved information to infer sensitive information about individual records in the raw data. In light of the increasing number of real-world applications of (privacy-preserving) synthetic data tools [22], [26], [62], the question how to *quantitatively assess* how much a strategic adversary might infer from a synthetic dataset becomes an urgent problem to solve.

As outlined in Section I, previous approaches to evaluate the privacy properties of synthetic data do not provide a rigorous assessment of an adversary’s inference power given access to a synthetic dataset: Evaluation methods based on similarity tests between real and synthetic records [28], [40], [41] do not measure the robustness of synthetic data against inference attacks. Formal privacy guarantees for the model training process [42], [43] do not allow to draw direct conclusions about the protection against inference attacks on the model’s output. Studies on the vulnerability of generative models against model-specific extraction attacks [45]–[47] assume that the adversary either has query access to the trained model or direct access to its parameters.

In this section, we introduce a novel evaluation framework to assess (claims about) the privacy properties of synthetic data publishing. The framework provides data holders and researchers with a tool to evaluate whether a synthetic dataset generated by a statistical model trained on a sensitive dataset protects the confidentiality of individual records in the raw data. In contrast to previous evaluation approaches, the framework allows to directly assess whether publishing a *single copy* of synthetic data in place of the raw data reduces the risk of private information leakage through *inference attacks*. Due to its modular nature, the framework is not limited to a unique threat model [63] but enables data holders to adapt the evaluation to any privacy concern specific to the data holder’s use case. In contrast to model-specific evaluation techniques [64], our framework treats the data generating mechanism as a complete black-box and evaluates the privacy risks of synthetic data *publishing* rather than an adversary’s inference power when given query access to a model [46], [63]. As the proposed evaluation method is independent of the generative model, it can be applied to synthetic data generated by models trained without any explicit privacy protection or trained under formal privacy guarantees [42], [43].

In Sections V and VI, we instantiate our framework to evaluate whether synthetic data generated by five common generative model types indeed provides “absolute privacy” [26] with

respect to these attacks. Our experimental results demonstrate that, across the board, synthetic data does not fulfil its high promises but rather that *previous evaluation approaches did substantially underestimate the privacy risks of synthetic data publishing*.

A. Framework formalisation

In this section, we provide a formal description of the proposed evaluation framework. In short, the framework measures the privacy gain of publishing a synthetic dataset in-place of the raw data with respect to a specific privacy concern. Each concern is modelled as a *privacy adversary* that targets an individual record and aims to infer a secret about this record. To evaluate *privacy gain*, the framework is instantiated under the pre-defined threat model and outputs an estimate about how much publishing the synthetic data instead of the raw data reduces the *privacy loss* of a chosen target record under this threat model.

Throughout this paper, we use $X \sim \mathcal{D}^n$ to refer to a generic dataset of size n sampled from an unknown distribution $\mathcal{D}$ over the data domain of the population $\mathcal{R}$. X could either be the raw data R , or a sanitised or synthetic version of R .

Adversarial models. In our framework, we measure the privacy gain of publishing S instead of R with respect to a *specific privacy concern* about the potential disclosure of a secret t_s associated with an individual target record $\mathbf{t} \in \mathcal{R}$. We model a privacy concern as an adversary $\mathcal{A}_{\mathbf{t}}(X) : X \rightarrow \hat{t}_s$ who, given access to a dataset X , optimises her strategy $\mathcal{A}_{\mathbf{t}}()$ to produce an estimate $\hat{t}_s$ of the secret t_s of a target record $\mathbf{t}$. We denote the adversary’s training procedure that outputs the function $\mathcal{A}_{\mathbf{t}}()$ as $\text{AdvTrain}(\mathbf{t}, \dots)$. The internals of this procedure depend on the privacy concern and the specific threat model. We provide example implementations of this procedure to address concerns regarding the risk of linkability and attribute inference in Sections V and VI.

Privacy loss. We first define the *privacy loss* for an individual record $\mathbf{t}$ under an adversary $\mathcal{A}_{\mathbf{t}}()$ that is trying to infer secret t_s linked to record $\mathbf{t}$ given access to dataset X .

Definition 1 (Individual Record Privacy Loss): Let $X \sim \mathcal{D}^n$ be a dataset of size n , $\mathbf{t}$ a random record from $\mathcal{R}$, and t_s a secret associated with $\mathbf{t}$. We define the *individual record privacy loss* $\text{PL}_{\mathbf{t}}(X)$ for target $\mathbf{t}$ under adversary $\mathcal{A}_{\mathbf{t}}(X)$ as

$$\text{PL}_{\mathbf{t}}(X) \triangleq \mathbb{P}[\hat{t}_s = t_s | X] - \mathbb{P}[t_s], \quad (1)$$

where $\hat{t}_s$ is the guess output by $\mathcal{A}_{\mathbf{t}}(X)$, $\mathbb{P}[\hat{t}_s = t_s | X]$ is the adversary’s success probability given X , and $\mathbb{P}[t_s]$ is her prior distribution over t_s . We write $\mathbb{P}[\mathcal{A}_{\mathbf{t}}(X) = t_s]$ as a shorthand for $\mathbb{P}[\hat{t}_s = t_s | X]$ under $\mathcal{A}_{\mathbf{t}}()$. *Bris:* The notation is never used, so no need for this. If t_s is a continuous attribute and $\mathbb{P}[t_s | X]$ defines a probability density function, we calculate $\hat{t}_s = t_s$ as $\hat{t}_s \in [t_s - \epsilon, t_s + \epsilon]$ for a small ϵ .

In this definition, the adversary’s prior and posterior probabilities of success take values in $[0, 1]$, and thus $\text{PL}_{\mathbf{t}}(X)$ ranges between $[-1, 1]$. $\text{PL}_{\mathbf{t}}(X) = 1$ implies that access to X substantially improves the adversary’s chance of making a

correct guess with respect to her prior. Negative privacy loss values, $\text{PL}_t(X) < 0$, indicate that observing X confuses the adversary, i.e., her success rate is smaller than her prior.

Privacy gain. We define the *privacy gain* for an individual record t when publishing dataset X_1 compared to X_2 as:

Definition 2 (Individual Record Privacy Gain): Let $X_1 \sim \mathcal{D}^n$ and $X_2 \sim \mathcal{D}^m$ be two datasets from the same domain, and t a random record from $\mathcal{R}$. The *privacy gain* for t of publishing X_2 compared to X_1 is defined as:

$$\text{PG}_t(X_2, X_1) \triangleq \frac{\text{PL}_t(X_1) - \text{PL}_t(X_2)}{2}, \quad (2)$$

where $\text{PL}_t(X_1)$ and $\text{PL}_t(X_2)$ denote t 's privacy loss under $\mathcal{A}_t(\cdot)$ given X_1 and X_2 , respectively. The multiplicative factor $1/2$ normalises PG to the interval $[-1, 1]$.

Under most threat models, including our example instantiations of the risk of attribute inference (Section V) and linkability (Section VI), the adversary's prior over the target's secret $P[t_s]$ is independent of the published dataset X . In this case, the record privacy gain can directly be calculated as

$$\text{PG}_t(X_2, X_1) = \frac{P[\hat{t}_s = t_s | X_1] - P[\hat{t}_s = t_s | X_2]}{2}. \quad (3)$$

We use this definition to measure the privacy gain of publishing a synthetic dataset S sampled from a generative model trained on R instead of R itself. $\text{PG}_t(S, R) > 0$ indicates that the synthetic dataset increases the target's privacy with respect to R , while $\text{PG}_t(S, R) < 0$ indicates that the synthetic dataset actually leaks more information about the target to the adversary than R . The extreme case of $\text{PG}_t = 1$ indicates that publishing S in place of R successfully prevents a breach of the target's privacy: While publishing R would incur a privacy loss of $\text{PL}_t = 1$ to target record t , the synthetic dataset S successfully mitigates the risk modelled by $\mathcal{A}_t(\cdot)$. Vice versa, $\text{PG}_t = -1$ implies that the synthetic dataset S maximises the target's privacy risk while R provides maximum protection against this adversary. If $\text{PG}_t = 0$ publishing S is equivalent to publishing R in terms of privacy for record t .

B. Framework implementation and practical evaluation

We implemented the building blocks of the privacy evaluation framework as a Python library [65]. Appendix B describes the architecture and modules of our open-source implementation that will allow others to replicate our findings or apply the framework to other data generating algorithms not described here.

In Sections V and VI, we use our implementation and instantiate the framework to assess the expected privacy gain for a set of randomly chosen target records under five generative model training algorithms (see Section IV-A). The expected record privacy gain provides a useful measure to not only assess the overall expected privacy gain for a dataset R but also to analyse whether privacy gain is distributed uniformly across target populations.

In our experiments, we use the *expected record privacy gain* to compare the privacy protection awarded by different generative models across datasets and target populations.

Definition 3 (Expected Record Privacy Gain): A target t 's expected privacy gain under a fixed model training algorithm $\text{GM}(\cdot)$ can be estimated as

$$\overline{\text{PG}_t} = \mathbb{E}_{\substack{R \sim \mathcal{D}_{\mathcal{R}}^n \\ f(R) \sim \text{GM}(R) \\ S \sim \mathcal{D}_{f(R)}^m}} [\text{PG}_t(S, R)] \quad (4)$$

where R is the input dataset of size n , PG_t is t 's privacy gain from publishing S instead of R , and S a synthetic dataset generated by a generative model $f(R)$ that was obtained by running $\text{GM}(R)$.

This definition makes it apparent why it is not straightforward to assess how much privacy one can expect to gain from publishing a synthetic dataset in place of the raw data. Such an assessment requires to take into account the randomness of the generative model training $f(R) \sim \text{GM}(R)$, the randomness of the synthetic data generation process $S \sim \mathcal{D}_{f(R)}^m$, and the randomness over possible datasets $R \sim \mathcal{D}_{\mathcal{R}}^n$. Furthermore, a data holder who seeks to assess whether publishing a synthetic dataset S sufficiently protects the privacy of all individuals needs to evaluate the expected privacy gain over all possible targets $t \in \mathcal{R}$.

IV. EXPERIMENTAL EVALUATION SETUP

For our experimental evaluation, we integrated five existing generative model training algorithms in our framework and tested them on three commonly used benchmark datasets following the procedure described in Section IV-C.

A. Generative models

We implemented three generative models without explicit privacy protection and two models designed to offer DP. We chose models relevant to the tabular data sharing use case and to cover a wide range of model architectures. We also considered their computational feasibility and whether a working implementation was available. Table III in Appendix C summarises our parametrisation of these models.

IndHist. We adopted a simple independent histogram model in which each attribute in the tabular input data is modelled independently from Ping et al. [33]. The IndHist training algorithm extracts marginal frequency counts from each data attribute and generates synthetic datasets S by sampling from each learned histogram independently. Continuous attributes are binned and the number of bins `nbins` is a model parameter.

BayNet. Bayesian networks capture correlations between attributes by factorising the joint data distribution as a product of conditionals. The degree of the network model is a model parameter. The trained network provides an efficient way to sample synthetic records from the fitted distribution (see Zhang et al. [34] for details). We use the GreedyBayes implementation provided by Ping et al.'s DataSynthesizer [33].

PrivBay. PrivBayes [34] is a differentially private Bayesian network model. Both, the Bayesian network and the conditional distributions, are learned under ε -differentially private algorithms. The resulting ε -differentially private model enables to sample synthetic records without any additional privacy budget cost. We use the GreedyBayes procedure provided by Ping et al. [33] to train a differentially private version of BayNet.

CTGAN. CTGAN [29] uses mode-specific normalisation of tabular data attributes to improve the approximation of complex distributions through GANs. CTGAN further uses a conditional generator and training-by-sampling to get better performance on imbalanced datasets.

PATEGAN. PATE-GAN builds on the Private Aggregation of Teacher Ensembles (PATE) framework [66] to achieve DP for GANs [67]. PATE-GAN replaces the discriminator’s training procedure with the PATE mechanism. The trained model provides DP with respect to the discriminator’s output.

B. Datasets

We include three tabular datasets, commonly used in the machine learning (ML) literature, in our experimental evaluation. Tabular datasets are the most relevant data type in the synthetic data publishing case. Two datasets contain financial data, and the third one contains health data:

Adult [68]. The Adult dataset contains information from 45,222 individuals extracted from the 1994 US Census database. Each entry consists of 15 attributes among which 6 are continuous attributes and 9 are categorical attributes.

Texas [69]. The Texas Hospital Discharge dataset is a large public use data file provided by the Texas Department of State Health Services. The dataset we use consists of 50,000 records uniformly sampled from a pre-processed data file that contains records from 2013. We retain 18 data attributes of which 11 are categorical and 7 continuous attributes.

German [70]. The original German Credit dataset contains data from 1000 individuals with 20 attributes. We use a pre-processed version of the dataset [71] which consists of 10 attributes for each individual of which 3 are continuous and 7 are categorical.

C. Experimental setup

We describe the experimental procedure we follow in Sections V and VI to empirically evaluate the privacy gain of the five generative models described above regarding the risks of attribute inference and linkability using our framework implementation [65].

Worst-case evaluation. We evaluate the individual record privacy gain of a crafted target $\mathbf{t}_c \sim \mathcal{D}_{R^*}$. This crafted target is designed to be an outlier, and that represents a very vulnerable target record due to its unusual distribution of attribute values. We craft this target by creating a distribution $\mathcal{D}_{R^*}$ as follows: For each categorical attribute, we assign probability 1 to the least frequent category in the original dataset R . For each

TABLE I: Experimental setup parameters

Dataset	n	m	n_T	n_R	n_S	n_A	k
Adult	1,000	1,000	50	25	100	10,000	10
Texas	1,000	1,000	50	25	100	10,000	10
German	300	300	50	25	100	500	10

numerical attribute, we assign probability 1 to the maximum attribute value.

Average case evaluation. We run n_R independent tests in which we evaluate the individual record privacy gain for a fixed set of n_T target records: $T = (\mathbf{t}_1, \dots, \mathbf{t}_{n_T})$. These records are randomly chosen from the dataset R . In each run, we repeat the following procedure: (1) We sample a new target training set $R^t \sim \mathcal{D}_R^n$ and use it as input to obtain five trained target models $f(R^t)$. (2) From each model $f(R^t)$, we sample n_S synthetic test datasets $S_1^t, \dots, S_{n_S}^t$ of size m . (3) For each $\mathbf{t} \in T$, we run an adversary $\mathcal{A}_{\mathbf{t}}()$ on each synthetic dataset to obtain n_S independent estimates of the target record’s privacy loss $\text{PL}_{\mathbf{t}}(S_i)$. (4) Finally, we estimate a record’s privacy gain $\text{PG}_{\mathbf{t}}$ for dataset R^t and trained model $f(R^t)$ as the average gain across all n_S synthetic samples from $f(R^t)$: $\text{PG}_{\mathbf{t}} = \frac{\text{PL}_{\mathbf{t}}(R^t) - \overline{\text{PL}_{\mathbf{t}}(S)}}{2}$.

Table I summarises the parameters used in the evaluation. n_A and k are parameters specific to the adversary that implements the risk of linkability described in Section VI. We set these parameters based on a series of experiments in which we varied each parameter to measure its impact on privacy gain and chose each value to present a worst-case bound on privacy.

V. SYNTHETIC DATA DOES NOT MITIGATE ATTRIBUTE INFERENCE RISKS

We model the risk of attribute inference as an adversary that aims to infer a sensitive attribute value for a partially known target record.

Previous work on attribute inference. The risk of attribute inference attacks on (sanitised versions of) sensitive datasets has been modelled in a variety of ways. In the context of data sanitisation techniques, such as generalisation [61] or partial imputation [27], [72], attribute inference is commonly modelled as record linkage. These works assume an adversary who has partial knowledge of a target record and aims to *link* the target record to a sensitive dataset to reconstruct a missing sensitive value. As such attacks rely on literal record linkage, they do not provide a useful adversarial model for synthetic data publishing in which no direct link between real and synthetic records exists. A more applicable adversarial setting is presented by works on attribute inference from social network data [73]–[79]: given access to the social graph of users and their preferences, the adversary aims to infer individual users’ traits that are not necessarily contained in the original data, e.g., a user’s political views [73]. Previous research on attribute inference attacks on ML models work with a different definition of what constitutes a privacy breach: The output of

a trained model should not reveal any information that does not relate to the model’s original task (defined by its training objective) [80], [81] or that is considered “unexpected” [44]. In the case of generative models trained on tabular microdata, however, a concise definition of the model’s task often does not exist.

In this work, we introduce a new notion of attribute inference for the context of synthetic data publishing that combines ideas from these two lines of work, concretely the work on record linkage by Machanavajjhala et al. [61] and the correlation-based inference by Zheleva and Getoor [73].

A. Attribute inference on synthetic datasets

Adversarial model. We formalise the risk of attribute inference as a regression problem in which an attacker learns to predict the value of an unknown sensitive attribute from a set of known attributes given access to a (raw or synthetic) dataset.

Definition 4 (Attribute Inference Attack): Let $\mathbf{t}$ be a target record randomly chosen from $\mathcal{R}$. Let $\mathbf{t}_k = (r_1, \dots, r_k)$ be the k attributes of record $\mathbf{t}$ known to an adversary $\mathcal{AI}_t()$ and t_s the target’s sensitive attribute. An attribute inference attack on target $\mathbf{t}$ based on background knowledge $\mathbf{t}_k$ is a mapping from dataset X to a guess about the target’s secret: $\mathcal{AI}_t(X) \rightarrow \hat{t}_s$. The dataset X can be (a sanitised version of) the raw data R or a synthetic dataset S .

In the following, we assume that the sensitive attribute t_s takes values in the continuous domain $\mathbb{R}$. We note, however, that our attack could easily be extended to categorical sensitive attributes, or to guess multiple sensitive values simultaneously, by replacing the attack’s predictive model with a multi-label classifier.

As an illustrative example, we instantiate an attribute inference attack based on a simple linear regression model. While other, more sophisticated predictors could be used, we find that already this simplistic approach results in a powerful attack (see Section V-D).

To learn a mapping from known to sensitive attributes, the adversary splits the dataset into two parts. One part represents all attributes for which the attacker knows the target’s values: R_k , a real-valued matrix with n rows each containing the k known attribute values. The other part contains the sensitive attribute values of all records in R : $\mathbf{r}_s$, a vector of size $n \times 1$.

The linear regression attack models the relationship between the sensitive attribute values in $\mathbf{r}_s$ and the attributes in R_k as a linear relationship with coefficients $\mathbf{w}$, and treats the records in R as n i.i.d. samples:

$$\mathbf{r}_s = R_k \mathbf{w} + \epsilon, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (5)$$

During training, the adversary takes the dataset R , splits it into a feature matrix R_k and target variable $\mathbf{r}_s$, and uses maximum likelihood estimation to obtain a set of regression coefficients $\mathbf{w}_R = \max_{\mathbf{w}} \mathbb{P}[\mathbf{r}_s | R_k, \mathbf{w}]$. Algorithm 1 implements the training procedure for this adversary. It outputs an attack function $\mathcal{AI}_t(R)$ that, without any further input, returns

the result of applying the learned linear coefficients $\mathbf{w}_R$ to the target record $\mathbf{t}_k$.

Algorithm 1 AITrain

Input: $R, \mathbf{t}_k, k, s$
Output: $\mathcal{AI}_t(R)$
1: $R_k, \mathbf{r}_s \leftarrow R$
2: $\mathbf{w}_R \leftarrow \text{MLE}(R_k, \mathbf{r}_s)$
3: $\mathcal{AI}_t(R) \leftarrow \mathbf{t}_k \mathbf{w}_R$

During the attack, the adversary uses the learned coefficients, $\mathbf{w}_R$, to predict a target’s secret t_s from a partially known target record $\mathbf{t}_k$ according to:

$$\mathcal{AI}_t(R) \triangleq \mathbf{t}_k \mathbf{w}_R. \quad (6)$$

If instead of a raw dataset R , the adversary is given access to a synthetic dataset S , the adversary would obtain linear coefficients $\mathbf{w}_S$ and use them in Equation 6 to predict the target’s secret.

B. Privacy gain with respect to linkage risks

To evaluate the privacy gain (see Def. 2) for target $\mathbf{t}$ with respect to the risk of attribute inference, we train two different adversaries: $\mathcal{AI}_t(R)$ who obtains linear coefficients $\mathbf{w}_R$ from the raw data and $\mathcal{AI}_t(S)$ with $\mathbf{w}_S$ obtained from a synthetic dataset S sampled from a model trained on R .

We assume that the adversary’s prior over the target’s secret $\mathbb{P}[t_s]$ is independent of which dataset, R or S is published. We thus directly calculate the target’s privacy gain as the difference in the adversary’s success rate (see Equation 3) under $\mathcal{AI}_t(R)$ and $\mathcal{AI}_t(S)$. As we assume t_s is a continuous attribute, we calculate the adversary’s probability of success as $\mathbb{P}[\mathcal{AI}_t(X) \in [t_s - \epsilon, t_s + \epsilon]]$ for $\epsilon = 10^{-9}$.

The privacy gain PG_t with respect to attribute inference takes values in $[-0.5, 0.5]$. $\text{PG}_t = 0.5$ indicates that the linear relationships between the known and sensitive attributes in R would allow an adversary to correctly predict a target’s secret t_s ($\mathbb{P}[\mathcal{AI}_t(R) = t_s] = 1$) while the synthetic dataset S reduces the adversary’s chance of success to $\mathbb{P}[\mathcal{AI}_t(S) = t_s] = 0$. A negative gain $\text{PG}_t < 0$ indicates that publishing the synthetic data S actually improves the adversary’s chance to correctly infer t_s from $\mathbf{t}_k$ over publishing R .

Algorithm 2 AIGain

Input: $\text{GM}(), \text{AITrain}(), R, \mathbf{t}, k, s, n_s$
Output: PG_t
1: $\mathbf{t}_k, \mathbf{t}_s \leftarrow \mathbf{t}$
2: $\mathcal{AI}_t(R) \leftarrow \text{AITrain}(R, \mathbf{t}_k, k, s)$
3: $\mathbf{f}(R) \sim \text{GM}(R)$
4: **for** $i = 1, \dots, n_s$ **do**
5: $\mathbf{S}_i \sim \mathbf{f}(R)$
6: $\mathcal{AI}_t(\mathbf{S}_i) \leftarrow \text{AITrain}(\mathbf{S}_i, \mathbf{t}_k, k, s)$
7: $\text{PG}_t \leftarrow \frac{\mathcal{AI}_t(R) - \mathcal{AI}_t(S)}{2}$

Estimating record privacy gain. Algorithm 2 describes a procedure to estimate the record privacy gain with respect to attribute inference for a fixed target record and dataset. It

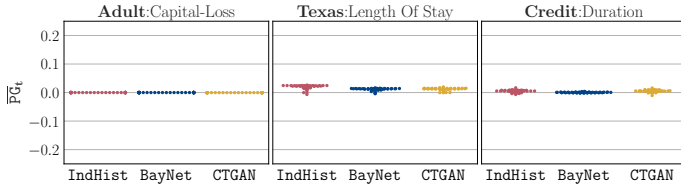


Fig. 1: Expected record privacy gain across 50 randomly chosen target records for different datasets, sensitive attributes, and generative models.

takes as input a model training procedure $GM()$, the adversary’s training algorithm $AI_{train}()$ the dataset R and target record t , attribute indices k and s , and the number of synthetic test datasets n_s .

We split the target record into the partial record known to the adversary, t_k , and the target’s secret t_s (line 1). We then run the adversary’s training procedure $AI_{train}()$ on the raw datasets R (line 2) and fit a generative model $f(R)$ to dataset R (line 3). We sample n_s synthetic datasets from $f(R)$ and train a privacy adversary on each of them (lines 4–6). Finally, we calculate the target’s expected privacy gain as the difference between the adversary’s chance of success given R and the average probability of success given S (line 7). We use $AI_t(R)$ as a shorthand to denote the adversary’s success probability $P[AI_t(R) = t_s]$. $\overline{AI_t}(S)$ is the average success probability of the adversary across all n_s synthetic samples.

C. Implementation

We implement the attribute inference adversary defined in Section V-A as an instantiation of the `PrivacyAttack` class (see Appendix B). We use a simple linear regression model provided by the `sklearn` library [82] to implement the adversary’s training. We centre all features extracted from the input data and fit a linear model without intercept. The model fits linear coefficients that minimise the root mean squared error between the observed and predicted target values. We then analytically calculate the adversary’s probability of success as: The simplicity of the model enables to analytically derive the attacker’s posterior distribution over the target’s secret given access to dataset R :

$$P[\hat{t}_s | R, t_k] = \mathcal{N}[t_k \mathbf{w}_R, \hat{\sigma}_R^2], \quad (7)$$

with variance $\hat{\sigma}_R^2 = \frac{1}{n-k} \sum_{i=1}^n (r_{s_i} - \mathbf{r}_{k_i} \mathbf{w}_R)^2$. To estimate the adversary’s probability of success when given access to synthetic dataset S , one can simply substitute R with S .

D. Experimental results

We evaluated the privacy gain with respect to the risk of attribute inference on three sensitive attributes randomly selected from each of the three datasets described in Section IV-B. We assessed privacy gain under all generative models described in Section IV-A using the experiment detailed in Section IV-C. The same experiment on other attributes yielded similar results.

Expected record privacy gain. In Fig. 1, we compare the expected record privacy gain PG_t (see Def. 3) for 50 randomly

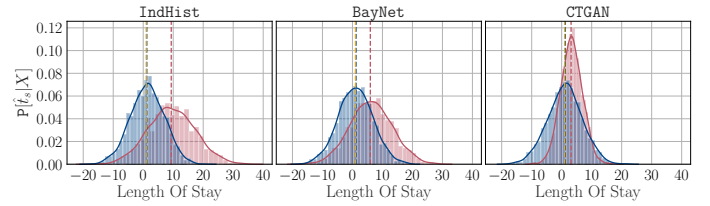


Fig. 2: Adversary’s distribution over the target’s secret after observing the raw data R ($-- P[\hat{t}_s | R]$) and synthetic data S ($-- P[\hat{t}_s | S]$). $--$ indicates the target’s true secret value.

chosen targets across 25 test runs (see Section IV-C). We fixed the set of target records and in every run we sampled a new training cohort R^t (see Section IV-C). The expected gain for each target t was estimated as the average gain across all target training sets R^t . Fig. 1 shows that the expected gain of publishing a synthetic dataset S in-place of the raw data R^t is close to 0 across datasets, sensitive attributes, generative models, and targets. For 70% of cases, the expected gain is $PG_t < 0.01$. Publishing S in place of the raw data R^t does not change the adversary’s probability of success for these targets. Only for attacks on the sensitive attribute Length of Stay in the Texas dataset we observe a slight variance in the expected gain. In this case, the expected gain for 66% of targets is larger than 0.02 for models trained using the IndHist algorithm, while 6% of target records have a small negative gain. If a CTGAN-trained model is used, the ratio of targets with a small positive gain decreases to 0%.

To illustrate what this variation means, we visualise in Fig. 2 the adversary’s posterior distribution over the secret of an example target in the Texas dataset for a fixed dataset R^t and sensitive attribute (Length of Stay). The blue ($--$) and red ($--$) curves represent the probability that the adversary guesses a particular value for the target’s secret after observing R^t or S sampled from a model trained on R^t , respectively. The yellow dashed line ($--$) indicates the target’s true secret. We observe that the adversary’s posterior after observing the raw data R^t is closely centred around the target’s true secret ($--$ curve). In comparison, the adversary’s guess is shifted away from the true value after observing synthetic data produced by an IndHist-trained model ($--$ curve, left) but less so if the synthetic data is sampled from a model trained through CTGAN ($--$ curve, right). This indicates that IndHist-trained models do not accurately preserve the linear relationships between known and sensitive attributes contained in the raw data that would allow an attacker to correctly infer the target’s secret. This result is not unexpected: The simple data representation learned by the IndHist training algorithm does not explicitly capture pairwise attribute correlations. The signal that our simple linear attack relies on is thus likely to get lost during model training.

A comparison of the correlation matrices of the raw and synthetic data drawn from the five generative models evaluated in this paper visualises this effect. Fig. 3 shows that synthetic data drawn from BayNet mimics closely the raw data’s correlation structure. Synthetic data generated through independent

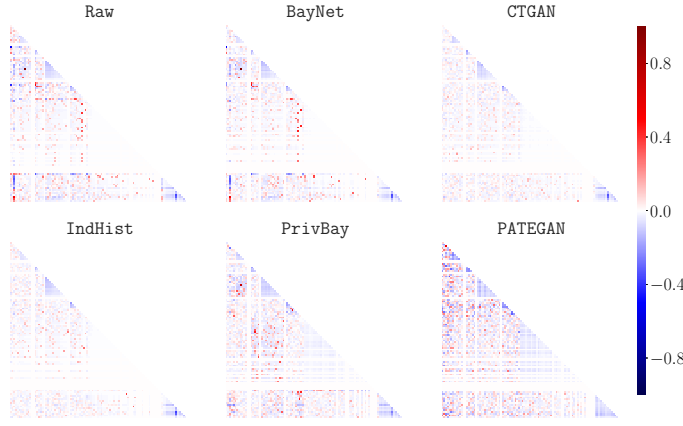


Fig. 3: Correlation matrices of a raw dataset sampled from the Adult dataset and synthetic datasets produced by five generative models trained on this data. Both differentially private models, PrivBay and PATEGAN were trained under $\varepsilon = 0.01$

attribute sampling from an IndHist-trained model only preserves strong correlations while weaker signals get lost. CTGAN seems to preserve some correlations but the representation is noisy.

No free lunch. These results demonstrate that also synthetic data is not, as promised by many, a privacy mechanism without utility trade-offs. Protection from attribute inference can only be achieved if the generative model does not accurately reproduce all correlations found in the raw data. If the correlations that enable an inference attack on the raw data are preserved in the model’s output, the gain of publishing the synthetic data in-place of the raw data is negligible. Any generative model, including privacy-preserving variants (see Section VII-A), are subject to this fundamental trade-off.

VI. SYNTHETIC DATA DOES NOT MITIGATE LINKAGE RISKS

In this paper, we model the risk of linkability as the risk of membership inference. A membership inference attack aims to determine whether a target record is present in a target model’s training set, i.e., the attacker aims to ‘link’ the target record to the training set. Successful linkage enables the adversary to learn sensitive information about the target record. For instance, linkage to a criminal records database can reveal the target has committed a particular crime.

Previous work on linkability and membership inference. Linkage attacks aim to establish a link between records concerning the same individual [48]. The threat of linkage attacks has been demonstrated, in theory and practice, for a large variety of data types: tabular micro-level datasets [59], [83], social graph data [8], [9], aggregate statistics [84], and statistical models [85]. Linkage attacks on tabular microdata usually intend to link a target record (connected to an identity) to a single record in a sensitive database from which direct identifiers have been removed. In effect, such linkage attacks re-identify records in the sanitised dataset [59], [83]. Membership inference attacks (MIAs) are linkage attacks which target

the output of statistical computations run on a sensitive dataset, such as aggregate statistics [84] or trained ML models [85]. ML-oriented MIAs have been extensively studied on predictive models, such as binary or multi-label classifiers [85]–[88]. Recently, this work has been extended to GANs [45], [46] and VAEs [47].

A. Membership inference on synthetic datasets

We build on existing MIAs on predictive models [85] to develop a *generic* black-box attack that is independent of the model architecture. We use this attack model to evaluate the risk that an attacker can link a target record to a sensitive dataset given synthetic data sampled from a generative model trained on this sensitive data.

Adversarial model. In contrast to previous attacks on GANs and VAEs [45]–[47], we assume an adversary that, instead of query access, only has access to a *single synthetic dataset output by the target model*. We define a privacy adversary $\mathcal{MLA}_t()$ that implements a generative model membership inference attack. Given a single synthetic dataset output by a generative model, this adversary produces a binary label that predicts whether a target record belongs to the model’s training set or not:

Definition 5 (Membership Inference Attack): Let $t_s = \mathbb{1}[t \in R^t]$ be a random variable that indicates $t \in R^t$ where R^t is the training set of a generative target model $f(R^t)$. A *membership inference attack* on target t given a synthetic dataset $S \sim \mathcal{D}_{f(R^t)}^m$ is a mapping $\mathcal{MLA}_t(S) \rightarrow \hat{t}_s$, $\hat{t}_s \in \{0, 1\}$.

Previous MIAs on predictive models range from “trivial” approaches, that simply use a correct or incorrect prediction to assign a membership label [88], to Bayes-optimal adversaries, that assume perfect knowledge of the underlying probability distributions [89]. We cast membership inference as a supervised learning problem [85]. Concretely, our membership inference adversary $\mathcal{MLA}_t()$ outputs a guess according to

$$\mathcal{MLA}_t(S) \triangleq \underset{t_s \in \{0,1\}}{\operatorname{argmax}} P[t_s|S]. \quad (8)$$

We estimate the posterior probability $P[t_s|S]$ using shadow training: we produce a set of generative shadow models trained on samples from a reference dataset R^a of size n_A with and without the target t , and train an attack model on synthetic datasets output by the shadow models labeled as *in* and *out*. As in [85], we assume that the adversary has access to the training algorithm $\text{GM}()$, and to a *reference dataset* $R^a \sim \mathcal{D}_{\mathcal{R}}^{n_A}$ that comes from the same distribution as the target model’s training data $R^t \sim \mathcal{D}_{\mathcal{R}}^n$ and may or may not overlap with R^t .

We describe the training procedure for the adversary $\mathcal{MLA}_t()$ in Algorithm 3. Given a reference dataset R , the adversary repeats the following steps k times. First, the adversary samples a training set R_i of size n from R (line 2) and trains a generative model (line 3). The adversary creates a set of n_S synthetic datasets of size m sampled from the trained model and assigns them the label 0 that indicates that the target record t was not present in the training data (lines 4–7). The adversary repeats the same procedure on the same

Algorithm 3 MIATrain

Input: $GM(), t, R, n, m, n_S, k$ **Output:** $MIA_t()$

```

1: for  $i = 1, \dots, k$  do
2:    $R_i \sim R^n$ 
3:    $f_i \sim GM(R_i)$ 
4:   for  $j = 1, \dots, n_S$  do
5:      $S_j \sim f_i$ 
6:      $S_{train} \stackrel{+}{\leftarrow} S_j$ 
7:      $l_{train} \stackrel{+}{\leftarrow} 0$ 
8:    $R'_i \leftarrow R_i \cup t$ 
9:    $f'_i \sim GM(R'_i)$ 
10:  for  $j = 1, \dots, n_S$  do
11:     $S_j \sim f'_i$ 
12:     $S_{train} \stackrel{+}{\leftarrow} S_j$ 
13:     $l_{train} \stackrel{+}{\leftarrow} 1$ 
14:  $MIA_t() \leftarrow \text{Classifier}(S_{train}, l_{train})$ 

```

training set, this time including the target, $R'_i = R_i \cup t$, and assigns the label 1 to the synthetic datasets to indicate that the target was present (lines 8-13). Finally, the adversary trains a classifier MIA_t which, given a synthetic dataset S , predicts the membership of the target record t in the training set of the generative model f that produced S (line 14).

B. Privacy gain with respect to linkage risk

In this section, we estimate privacy gain (see Def. 2) with respect to the risk of linkability that is implemented as a MIA. As in previous work [86], we assume an adversary that has a uniform prior over the target's membership in R ($P[t_s] = 0.5$). As the adversary's prior is equal in both cases regardless of whether the real or the synthetic data is published, the record privacy gain can be directly calculated as the difference between the adversary's chances of success given access to R and to S (see Equation 3). If the adversary has access to R , her probability to correctly guess whether the target record is present in R is 1 ($P[\mathcal{MLA}_t(R) = t_s] = 1$). Therefore, the record privacy gain for a target t with respect to linkability ranges between $PG_t = 0$, when publishing S leads to the same privacy loss as publishing R , and $PG_t = 0.5$, when the synthetic data perfectly protects the target from linkage. When publishing S does not hide nor give new information, the adversary's success probability to correctly infer membership is given by her prior knowledge. In this case $PG_t = \frac{1 - P[\mathcal{MLA}_t(S) = t_s]}{2} = \frac{1 - 0.5}{2} = 0.25$.

Estimating record privacy gain. In Algorithm 4 we describe a procedure to estimate the record privacy gain PG_t for a fixed target record and input dataset. This procedure takes as input a generative model training procedure $GM()$, a target training set R_{out}^t , the target record t , the size of the raw and synthetic datasets, n and m , the number of synthetic test sets n_S , a reference dataset R^a , and the number of shadow models k the adversary's training algorithm can use.

We first create an evaluation set of $2 * n_S$ synthetic datasets S_{test} . Half of the datasets are sampled from a generative model fitted to the raw dataset *without the target* R_{out}^t (lines

Algorithm 4 MIAGain

Input: $GM(), R_{out}^t, t, n, m, n_S, R^a, k$ **Output:** PG_t

```

1:  $f_{out} \sim GM(R_{out}^t)$ 
2: for  $i = 1, \dots, n_S$  do
3:    $S_i \sim f_{out}^m$ 
4:    $S_{test} \stackrel{+}{\leftarrow} S_i$ 
5:  $R_{in}^t \leftarrow R_{out}^t \cup t$ 
6:  $f_{in} \sim GM(R_{in}^t)$ 
7: for  $i = 1, \dots, n_S$  do
8:    $S_i \sim f_{in}^m$ 
9:    $S_{test} \stackrel{+}{\leftarrow} S_i$ 
10:  $MIA_t() \leftarrow \text{MIATrain}(GM(), t, R^a, n, m, n_S, k)$ 
11:  $PG_t \leftarrow \frac{1 - MIA_t(S_{test})}{2}$ 

```

1–4); half from a generative model fitted to the raw dataset *with the target record* in R_{in}^t (lines 5–9). We then train a privacy adversary $MIA_t()$ on target t using the model training procedure $GM()$ and a reference dataset R^a (line 10). We use this adversary to infer the presence of t in each of the synthetic datasets in S_{test} and we calculate the privacy gain for record t as the difference between the adversary's success probability under R^t ($P[MIA_t(R^t) = t_s] = 1$) and the average probability of success under S (line 11). In line 11, we write $MIA_t(S)$ as a shorthand for $P[MIA_t(S) = t_s]$ and $MIA_t(S_{test})$ for $\sum_{S_i \in S_{test}} MIA_t(S_i) / 2 * n_S$.

C. Practical considerations and implementation

Mounting a successful black-box MIA on a generative model is much more challenging than on a predictive model [46]. MIAs on predictive models leverage patterns in the confidence values that differ between two classes, members and non-members. MIAs on generative models need to identify the influence that *a single target record* has on the high-dimensional model's output distribution $\mathcal{D}_{f(R)}$. The non-deterministic output sampling process and the fact that the adversary only has access to a single output example from the trained model increases the difficulty of this task.

In our attack setup, the adversary needs to be able to distinguish between two distributions, $\mathcal{D}_{f(R)}$ and $\mathcal{D}_{f(R \cup t)}$, given a single observation $S \sim \mathcal{D}_{f(X)}^m$. Detecting the fine differences between $\mathcal{D}_{f(R)}$ and $\mathcal{D}_{f(R \cup t)}$, however, is next to impossible in the high-dimensional data domain of S . To reduce the effect of high-dimensionality and sampling uncertainty, the adversary can apply *feature extraction* techniques before training the classifier in line 14 of Algorithm 3. This means that, in practice, the adversary will not use Equation 8 but the following attack:

$$\mathcal{MLA}_t(F, S) = \underset{t_s \in \{0,1\}}{\operatorname{argmax}} P[t_s | F(S)], \quad (9)$$

where $F(S)$ denotes a function $F : S \rightarrow F$ that extracts a feature vector F of size n_F from a synthetic dataset S . Under this attack, the adversary effectively maps S back to a *lower dimensional feature space* and aims to detect the target's influence on these features. Whether the attack using feature

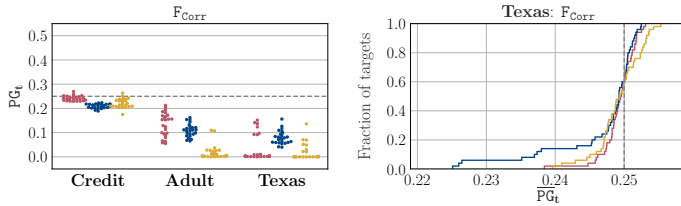


Fig. 4: *Left*: Record privacy gain for a crafted target across datasets and generative models. *Right*: Cumulative distribution of record privacy gain across randomly sampled target records for the `Texas` dataset under three generative models. Generative models are colour-coded as ● IndHist ● BayNet ● CTGAN.

set F is successful depends on two factors: First, whether the target’s presence has a detectable impact on any of the features in F , and second, whether the synthetic dataset has preserved these features from the raw data and hence preserved the target’s signal. We elaborate on these two conditions for attack success in Section VIII.

In our evaluation we implement the membership inference adversary $\mathcal{MIA}_t()$ as an instantiation of our framework’s `PrivacyAttack` class (see Appendix B). We leverage the object-oriented structure of the library to create multiple attack versions that share the same training procedure `MIATrain` but use different attack models and feature extraction techniques. We implement the adversary’s feature sets as feature extraction objects `FeatureSet`. Each `FeatureSet` takes in a synthetic dataset S of size $m \times k$ and outputs a vector of size $n_F \times 1$.

As feature extractors, we implemented a naive feature set with simple summary statistics F_{Naive} , a histogram feature set that contains the marginal frequency counts of each data attribute F_{Hist} , and a correlations feature set that encodes pairwise attribute correlations F_{Corr} .

As attack models, we implemented a Logistic Regression, Random Forests and K-Nearest Neighbours classifier (see Appendix B). All attack models yielded similar results with Random Forests performing best across datasets, generative models, and feature sets. In the next section, we focus on results obtained using the best performing attack model, Random Forests, and the attack feature set with the overall highest success rate F_{Corr} . In Section VIII, we discuss the influence of the attacker’s feature set on the expected privacy gain.

D. Experimental results

We evaluate the expected privacy gain with respect to the risk of linkability under all generative models described in Section IV-A, using the procedure described in Section IV-C. At the beginning of each experiment, we sampled a fixed reference dataset R^a from R and used it across all test runs and targets. The parameter n_A in Table I describes the size of the adversary’s reference dataset and k the number of shadow models trained by the adversary.

Worst-case privacy gain. We show in Fig. 4 *left* the estimated record privacy gain for a record crafted to represent a worst-case target, i.e., a target likely to be vulnerable to membership inference due to its unusual combination of attribute values

(see Section IV-C). We plot the estimated record privacy gain of this target for $n_R = 25$ test runs. In each run, we sample a new target training set R^t i.i.d from $\mathcal{R}$, and compute the estimated record privacy gain PG_t per target training set R^t as specified in Algorithm 4.

For the two bigger datasets (`Adult` and `Texas`), the estimated record privacy gain for a worst-case target consistently lies *well below the expected gain* of $\text{PG}_t = 0.25$ for all three generative models. For the `German` dataset, we observe that the estimated gain is much closer to the expected value of 0.25 which indicates that the adversary improves only slightly over her prior. This is likely due to the small size of this dataset. A generative model, when trained on such small amounts of data (< 1000 records per training set R^t) is unlikely to converge. Therefore, models (both targets and shadows) trained on small samples of the `German` dataset are not a robust representation of their input data. The attacker’s strategy to use shadow models to mimic the target model’s behaviour is hence less likely to succeed than on the two bigger datasets.

More worryingly, there exist training sets R^t for which the record privacy gain for the crafted target is *close to 0*. For those targets, publishing a synthetic dataset S instead of the raw dataset R^t would result in little to no additional privacy protection against linkage. In particular, for 68% of training sets from the `Texas` dataset the target’s gain is smaller than 0.01 for synthetic data produced by IndHist and CTGAN models, respectively. CTGAN-trained models provide the smallest overall gain with an expected privacy gain of $\overline{\text{PG}}_t = 0.02$. The low privacy gain in these cases indicates that the target’s presence has a detectable influence on some of the features used in the attack (F_{Corr}) and that the generative model clearly preserves this signal in its output.

Expected privacy gain across population. The previous results focus on a worst-case scenario in which the target is a rare outlier, and thus highly vulnerable to membership inference. In Fig. 4 *right*, we compare the expected record privacy gain $\overline{\text{PG}}_t$ (see Def. 3) for 50 randomly chosen targets across 25 test runs on the `Texas` dataset (see Section IV-C). We fix the set of target records and in every run we sample a new training cohort R^t . The expected gain for each target t is estimated as the average gain across all target training sets R^t .

As expected, the average gain for the randomly chosen targets is higher than the average gain of a worst-case target. However, we still find that for all three models the expected privacy gain is less than the random guess baseline ($\overline{\text{PG}}_t = 0.25$) for 60% of targets. These records are likely to be vulnerable to linkage independent of the distribution of records in the training set: Even when assessing the average gain across target training sets $R^t \sim \mathcal{D}_{\mathcal{R}}^n$ these records have a lower than expected gain. While in the worst-case CTGAN-produced synthetic data provides the least protection, BayNet-trained models are more likely to leave a randomly chosen target record vulnerable to linkage through an attack based on the correlations feature set. We obtained similar results on the

Adult dataset.

High variance, low predictability. Other than the low minimum gain achieved for some target records, we observe a high variance in the record privacy gain across training sets for a fixed record (Fig. 4 *left*) as well as a high variance in the average gain across target records for a fixed model training algorithm (Fig. 4 *right*).

A high variance in the record privacy gain across training sets for a fixed target record demonstrates that it is the combination of training set R^t and target record t that determines whether the target has a detectable influence on the data features preserved by the model. This means that it is not possible to predict privacy gain for a particular target record without fixing the entire training set.

A high variance in the average privacy gain across target records indicates that the model tends to amplify the signal of some records in the training set while it actually hides the presence of others, i.e., it gives disparate protection. Experiments with attack features other than the correlations set F_{corr} show that the variance in privacy gain provided by CTGAN is highest ranging from 0.04 to 0.48 across targets (see Appendix D). We further investigate what causes this high variance in Section VIII-A.

VII. DIFFERENTIALLY PRIVATE MODEL TRAINING DOES NOT INCREASE PRIVACY GAIN

In previous sections, we demonstrate that synthetic data drawn from three common generative models provides little gain with respect to an adversary that aims to infer a sensitive attribute value for a partially known target record and fails to provide robust protection against linkage attacks. This raises the question whether generative models explicitly designed to protect the privacy of inputs through privacy-preserving training techniques suffer from the same shortcomings. In this section, we extend our evaluation to two model training algorithms, PrivBay and PATEGAN, that provide formal privacy guarantees and assess the expected privacy gain with respect to the two adversaries introduced in Section V and VI.

Related work on differentially private synthetic data generation. Recent research introduces privacy-preserving generative model training based on the DP model [34], [43], [67]. These models aim to protect the privacy of the training data by providing formal privacy guarantees for *the lower-dimensional approximation of the full-dimensional data distribution*. Synthetic datasets drawn from the differentially private model preserve these privacy guarantees under the post-processing guarantee [12].

The model training algorithm PrivBay, for instance, learns a differentially private Bayesian network that approximates the data distribution via the exponential mechanism and computes the conditionally independent marginals in the subspaces of the Bayesian network via the Laplace Mechanism [34]. Thus, the trained model guarantees that neither the attribute correlations represented by the network nor the marginal histograms allow an attacker to infer the presence of a target record in

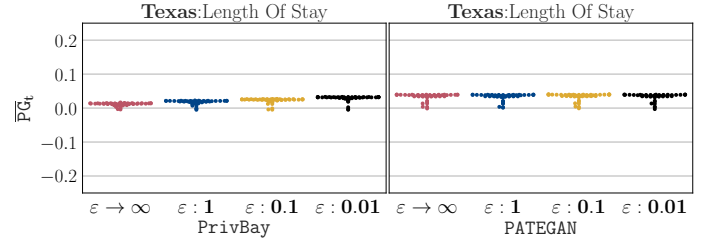


Fig. 5: Expected record privacy gain across 50 randomly chosen target records under PrivBay- and PATEGAN-trained models for four different privacy parameter settings. $\epsilon \rightarrow \infty$ indicates no noise addition during training. Data shown for the sensitive attribute Length of Stay in the Texas dataset.

the model’s training data. Similarly, PATEGAN ensures that the decisions of the discriminator network are not affected by the target’s presence by more than the defined ϵ bound [67].

A. Experimental results

We used the same attacks and experimental setup as in previous sections to empirically evaluate whether models trained under DP increase the expected record privacy gain with respect to the two major privacy concerns investigated in this work.

Attribute inference. In Fig. 5, we show the expected record privacy gain across 50 randomly chosen target records under PrivBay- and PATEGAN-trained models for four different privacy parameter settings. We show results for an attack that aims to infer the sensitive attribute Length of Stay in the Texas dataset but obtained similar results for all other attributes we investigated. For PrivBay, differentially private training results in a slight increase in privacy gain compared to an equivalent model without privacy protection ($\epsilon \rightarrow \infty$). The fraction of targets with $\overline{PG_t} > 0.02$ increases from 46% to 92% when the privacy parameter ϵ decreases from 1 to 0.01. However, even under differentially private training with $\epsilon = 0.01$ the maximum gain achieved by any one target is $\overline{PG_t} = 0.03$ which is a very moderate decrease in the attack’s success rate. The differentially private training through PATEGAN *does not* have any effect on the target’s privacy gain. Neither the maximum gain ($\overline{PG_t} = 0.04$) nor the fraction of targets with a gain above 0.02 (88%) changes with decreasing ϵ values.

Both models, PrivBay and PATEGAN, when trained under strong DP guarantees $\epsilon = 0.01$, provide a small gain in privacy over the non-private models investigated in Fig. 1. Fig. 3 again shows that this small gain in privacy comes at a cost in utility. While the vanilla Bayesian network model, with a maximum gain of 0.016, accurately replicates the correlation structure of the raw data, differentially private learning leads to a much noisier representation. For target records whose sensitive value could only be inferred based on the perturbed correlations, this results in an increase in privacy gain to a maximum of 0.03 for PrivBay and to 0.04 for PATEGAN. For analysts trying to draw insights, however, this reduces the synthetic data’s utility.

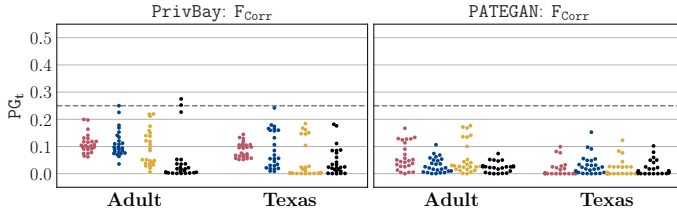


Fig. 6: Record privacy gain for a crafted target under PrivBay and PATEGAN-trained models for four different privacy parameter settings. The ϵ values are colour-coded as $\bullet \epsilon \rightarrow \infty$ $\bullet \epsilon = 1$ $\bullet \epsilon = 0.1$ $\bullet \epsilon = 0.01$

$\epsilon \rightarrow \infty$ indicates no noise addition during training. Data shown a Random Forests attack using the F_{Corr} feature set.

Membership inference. In Fig. 6, we show the estimated record privacy gain for a worst-case target under PrivBay and PATEGAN. As in previous experiments we observed that the German dataset does not provide enough data to obtain good model fits, we only present results for the Texas and Adult dataset. *Neither* differentially private training leads to an increase in average record privacy gain with respect to the risk of linkability. For instance, for 44% and 28% of training sets R^t from the Adult dataset the target’s estimated gain is close to 0 ($\text{PG}_t < 0.01$) for synthetic data drawn from a PrivBay and PATEGAN model with $\epsilon = 0.01$, respectively.

This implies that synthetic data drawn from either privacy-preserving model carries through enough information about the target’s presence for an attacker to successfully identify the preserved features and infer the target’s secret. This is especially surprising in the case of PrivBay under an attack focusing on the correlations feature set. The DP guarantee of the PrivBay procedure should ensure protection against linkage attacks on these features. Even more surprisingly, the expected gain of a worst-case target under this model even *decreases* as the privacy parameter decreases for both datasets (from $\overline{\text{PG}}_t = 0.11$ for $\epsilon = 1$ to $\overline{\text{PG}}_t = 0.04$ for $\epsilon = 0.01$ in the Adult dataset, statistically significant with $p < 0.0001$). We further investigate this finding in Section VIII-B.

No (free) gain. We draw two main conclusions from these experiments. First, differentially private noise addition may hide some of the linear relationships that enable attribute inference attacks. Hiding these correlations, however, also reduces the data’s utility. Second, privacy guarantees for the lower-dimensional representation of the input data learned by differentially private models *do not necessarily extend to all features of the high-dimensional synthetic data* sampled from these models. Thus, these models do not provide robust protection against linkage attacks on their outputs.

VIII. WHY GENERATIVE MODELS ARE POOR PRIVACY MECHANISMS

In previous sections, we demonstrate that synthetic data does not provide a magical, trade-off free mechanism for privacy-preserving data publishing. Any dimensions in the synthetic data that closely mimic raw data features increase the

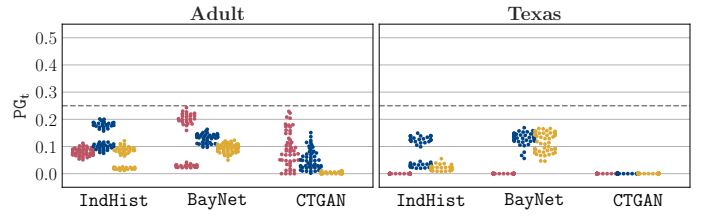


Fig. 7: Record privacy gain for a crafted target and fixed training set for different generative models under a Random Forests attack on varying feature sets. Feature sets are colour-coded as $\bullet F_{\text{Naive}}$ $\bullet F_{\text{Hist}}$ $\bullet F_{\text{Corr}}$

utility of the data but simultaneously allow targeted inferences about individual training records. This is neither a new nor a surprising insight and does not, per-se, hinder the use of generative models as privacy mechanisms.

A much more concerning observation, however, is that the privacy gain of publishing a synthetic dataset in-place of the raw data with respect to the risk of linkability varies substantially depending on the target record, input dataset, and model training algorithm. While some target records are protected from linkage attacks, other records receive close to zero protection and which records remain vulnerable depends on the dataset and the chosen model (see Section VI-D).

In this section, we demonstrate that, due to the stochastic nature of generative models and the high-dimensionality of the model’s output, it is *impossible* to give any guarantees about the level of protection generative models provide. We show that even differentially private model training can not result in robust protection and that therefore generative models should not be considered effective privacy mechanisms.

A. Unpredictable feature preservation

In Section VI-D, we observe that the record privacy gain for a worst-case target has high variance. This variance depends on two factors: the training set distribution and the generative model type. Here, we show that even when we fix both these factors, the entire training set and the model training algorithm, the record privacy gain remains unpredictable.

We conduct an experiment in which we fix a target record $\mathbf{t}$ and sample a fixed training set R^t of size n from R (see Table I for parameter details). For each dataset, we run 25 test runs in which we take *the same* dataset R^t and train two models, with and without the target: $f(R^t \cup \mathbf{t}) \sim \text{GM}(R^t \cup \mathbf{t})$ and $f(R^t) \sim \text{GM}(R^t)$. We sample multiple copies of synthetic data from both models and assess the target’s privacy gain PG_t under a Random Forests attack on different feature sets.

We show the results of this experiment in Fig. 7. *Even with the entire training cohort fixed and under the same model training algorithm*, the target’s estimated privacy gain varies across model training runs and across attack feature sets. For instance, for synthetic data produced by CTGAN models trained on the Adult dataset, the record privacy gain PG_t varies between 0.0 and 0.22 under the same attack (F_{Naive}) and the same training set. The target’s expected gain $\overline{\text{PG}}_t$ decreases from 0.09 to 0.004 if the attacker extracts correlations ($\bullet F_{\text{Corr}}$)

instead of simple summary statistics ($\bullet$ F_{Naive}) (statistically significant with $p < 10^{-12}$).

This experiment shows that the target’s privacy gain depends on a third factor that is not under the data publisher’s control: the feature set the adversary chooses. The level of protection a generative model provides depends on how much information the model’s output leaks about the features targeted by the attack. This means that we can only predict privacy gain if we can predict what features a potential adversary will target, *and* if we know which data characteristics will be preserved. In practice, none of these factors is predictable. Synthetic data effectively preserves a large set of data characteristics which *are not constrained* to the features explicitly represented by the model.

For instance, recall that in Fig. 3 we observe that even synthetic data produced through independent attribute sampling from an IndHist-trained model preserves *strong correlations* contained in the raw data. This explains why attacks based on the correlations feature set on synthetic data produced by IndHist models have a high success rate ($\bullet$ points in Fig. 7). Non-parametric models, such as CTGAN, do not provide a parametric specification for the data’s density function, i.e., they do not preserve a pre-determined set of features. In fact, the features preserved by non-parametric CTGAN models are highly variable and strongly affected by small changes in the training data distribution. This results in a high variance in the privacy gain across attack feature sets, training sets, and target records (see Appendix D).

In summary, even simple statistical models, which approximate the data distribution as the composition of a pre-determined set of features (see Section II), do not give any guarantees about what features *other than the features in the model’s lower-dimensional approximation* will be preserved. This increases the potential inference power of an adversary who can freely choose which features to leverage in her attack. An adversary can leverage *any* features the target’s presence is likely to influence and the model’s output might replicate. This makes it impossible to predict the privacy gain of synthetic data publishing and generative models poor privacy mechanisms.

B. Differentially privacy only protects some features

Recall that our experiments show that even for small ϵ values a ML-based MIA is able to identify features in the model’s output that significantly change when the target is present in the input data (see Section VII-A). This is counterintuitive: differentially private model training should bound changes in the output when adding a single target record to the input data to a small ratio defined by ϵ . In this section, we show why differentially private model training fails to provide more robust protection against linkage.

A possible explanation for this failure is that the DP guarantee provided by these models only covers *the limited set of features* in the model’s lower-dimensional approximation [34]. Thus, an adversary may be able to identify features in the model’s high-dimensional output that are not covered by the

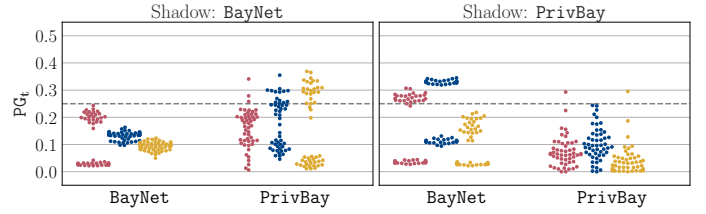


Fig. 8: Record privacy gain for a worst-case target for a fixed training set under two different adversary trained on different shadow models. The target and shadow models labelled PrivBay were trained under $\epsilon = 0.01$. Feature sets are colour-coded as $\bullet$ F_{Naive} $\bullet$ F_{Hist} $\bullet$ F_{Corr}

guarantee and that allow to infer the target’s membership. To test this hypothesis, we study whether DP provides protection against attacks that do not consider the defence and are thus more likely to target protected features. Fig. 8 shows the result of an experiment in which we fix a target record and training set and assess the success rate of an attack trained on shadows without formal privacy guarantees (Shadow: BayNet) on synthetic data drawn from a model with strong DP guarantees. We find that while such an attack under the F_{Corr} feature set is very successful on BayNet-generated data ($\overline{PG}_t = 0.09$), the target’s expected gain increases to 0.17 for synthetic data produced by a PrivBay target model with $\epsilon = 0.01$. This implies that an attack trained on shadows without privacy protection focuses on a subset of features that are (partially) protected in the output of a DP model. The PrivBay target model provides next to no protection against an adversary trained on differentially private shadow models (Shadow: PrivBay with $\overline{PG}_t = 0.04$). A feature importance analysis reveals that *there is no overlap between the top-10 most informative features* of the BayNet and PrivBay-trained attack classifiers.

This raises the question which of the features in the model’s high-dimensional output the PrivBay-trained attack exploits. We show in Appendix E that the DP guarantee for the structure of the model’s network does not extend to all possible pairwise attribute correlations in the synthetic data. Even though the lower-dimensional representation of the raw data learned under DP does not change depending on the target’s presence, some correlations in the model’s output are clearly affected and can be exploited by an adversary.

These experiments confirm why DP provides good protection against an attack trained on BayNet shadows: This attack likely learns to use changes in correlations directly caused by changes in the network structure depending on the target’s presence. Differentially private model training hides these correlations and the attack hence fails to reliably infer membership. The attack trained on PrivBay shadows instead identifies correlations that are influenced by the target’s presence but not explicitly modelled in the network and thus not covered by the DP guarantee. Even though the attacker has no guarantee that the targeted features will be preserved in the target model’s output, our results show that in practice this is a highly successful attack strategy.

IX. KEY TAKEAWAYS

Our evaluation framework enabled us to study the privacy gain provided by a wide variety of generative models, from simple statistical models to differentially private deep networks. Our results challenge the claim that synthetic data is a silver-bullet solution to the privacy problems of microdata publishing. We draw three important lessons from our experiments.

First, synthetic data does not, as promised by many [22], [26], [36]–[38], [53], enable data sharing without loss in utility or privacy. Any small gain in privacy comes at a cost in utility. Synthetic data that perfectly mimics the raw data offers useful information for analysis but simultaneously enables adversaries to infer sensitive attributes from partially known target records with high accuracy. If the synthetic data hides the patterns that enable the attack, it does so also from analysts.

Second, high-dimensional synthetic datasets often carry more information about their training data than what is captured by the generative model. Even synthetic data drawn from simple models often preserves a large number of raw data features. Adversaries are not constrained in their choice of feature set and therefore can use *any of the large number of data dimensions replicated in a model's output* to conduct their attack. This reminds of past failures of data anonymisation: “high-dimensional data is inherently vulnerable to de-anonymization” [90]. We find that the same lesson applies to the context of synthetic data publishing. In particular ML-based attacks have a high chance of identifying features that carry a target's signal and are likely to be preserved by a given generative model type. As it is not possible to predict neither what information synthetic carries nor what features adversaries target, it is impossible to predict the privacy gain of synthetic data publishing.

Third, our results show that DP provides little protection against ML-based inference attacks on high-dimensional data releases. The formal guarantees only cover the model's lower-dimensional approximation of the data but do not extend to the full-dimensional synthetic dataset drawn these models. Our findings imply that recent large-scale planned uses of synthetic [26] and differentially private synthetic data [22], [91], may need to be revisited.

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APPENDIX

A. Notation

Table II summarises frequently used notation.

TABLE II: Frequently used notation

Symbol	Meaning
$\mathcal{R}$	Population, a collection of data records
r_i	Real-valued data attribute
$\mathbf{r} = (r_1, \dots, r_k)$	Record, a real vector of k attributes
$\mathcal{D}_{\mathcal{R}} = \mathcal{P}[\mathbf{r}]$	High-dimensional probability distribution over the joint domain of all data attributes
$R = (\mathbf{r}_1, \dots, \mathbf{r}_n)$	Raw dataset of n real records
$\mathcal{D}_R = \mathcal{P}[\mathbf{r} R]$	Joint probability distribution defined by dataset R
$f(R) \sim \text{GM}(R)$	Generative model obtained running training algorithm $\text{GM}(X)$ on dataset X
$\mathcal{D}_{f(R)}$	Approximation of the joint distribution of dataset R through model $f(R)$
$S = (\mathbf{s}_1, \dots, \mathbf{s}_m)$	Synthetic dataset of m synthetic records
$\mathbf{t} = (r_1, \dots, r_k)$	Record targeted by the adversary
t_s	Secret of the target record
$\hat{t}_s = \mathcal{A}_{\mathbf{t}}(X)$	Guess on t_s when observing dataset X by an adversary trained to target record $\mathbf{t}$
$\mathcal{P}[\hat{t}_s = t_s X]$	The adversary’s probability of success given X
$\mathcal{P}[t_s]$	Prior probability over the secret
$\text{PL}_{\mathbf{t}}(X)$	Individual record privacy loss when publishing X
$\text{PG}_{\mathbf{t}}(X_2, X_1)$	Individual record privacy gain when publishing X_2 instead of X_1

B. Implementation of an open-source modular evaluation framework

We implemented the building blocks of the privacy evaluation framework as a Python library [65]. Fig. 9 shows

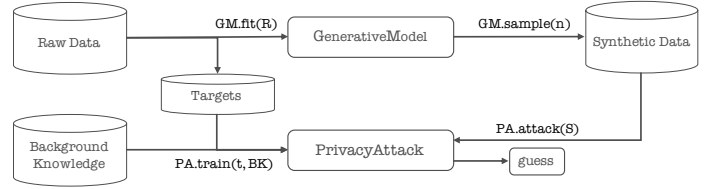


Fig. 9: Modules of the Python library that implements the privacy evaluation framework

an overview over the library’s main functions and objects. The library has two main classes: `GenerativeModels` and `PrivacyAttacks`. For both of them we define a parent class that determines the core functionality that objects of the class need to implement. A `GenerativeModel` needs to provide two main functions. `GM.fit()` is called with a raw training set R as input and implements the model’s training procedure (denoted as $f(R) \sim \text{GM}(R)$ throughout the paper). `GM.sample(m)` generates a synthetic dataset S of size m corresponding to $S \sim \mathcal{D}_{f(R)}^m$. The library is designed to enable easy integration of existing model training procedures. `GM.fit()` simply provides a wrapper for a training algorithm, exposing the appropriate API endpoints. To demonstrate this concept we include a number of existing generative models in our implementation of the evaluation framework.

`PrivacyAttack` objects have two functions: `PA.train` and `PA.attack`. `PA.train(t, BK)` trains the attack for a specific target record t on background knowledge BK . `PA.attack(S)`, takes a dataset S and outputs a guess about a secret value. In our implementation, we instantiate `PrivacyAttack` with two attacks, a membership inference adversary adapted from the work by Shokri et al. [85] (see Section VI); and a novel attribute inference attack (see Section V).

The library also provides procedures to estimate the record and generative model privacy gain.

Membership inference attack implementations We leveraged the modular structure of the library to implement multiple versions of the membership inference adversary presented in Section VI. As attack models we implemented:

LogReg. A binary logistic regression classifier regularised under the L2-norm.

RandForest. A random forest ensemble classifier with 100 estimators using the Gini impurity as splitting criterion.

KNN. A classifier using a k -nearest neighbours vote considering the $k = 5$ closest votes.

We implement the adversary’s feature sets as feature extraction objects `FeatureSet`. Each `FeatureSet` takes in a synthetic dataset S of size $m \times k$ and outputs a vector of size $n_F \times 1$. We compare the following feature extraction strategies:

F_{Naive}. The naive feature set computes the mean, median, and variance of each numerical attribute and encodes the number of distinct categories plus the most and least frequent category for each categorical attribute.

F_{Hist}. The histogram feature set computes the marginal distri-

bution of each data attribute. Numerical attributes are binned with configurable bin size and frequency counts are computed for categorical attributes. The number of bins per attribute is configured for each dataset independently and we list our experimental hyperparameters in Appendix C.

F_{Corr} . The correlations feature set encodes pairwise attribute correlations. Categorical attributes are dummy-encoded before the pairwise correlations between all attributes are computed.

F_{Ens} . The ensemble feature set combines the two feature vectors output by F_{Hist} and F_{Corr} and outputs a vector of length $n_{\text{Hist}} + n_{\text{Corr}}$.

C. Experimental hyperparameters

Generative model parameters. We experimentally evaluate the privacy gain of five common generative model training algorithms described in Section IV-A. For all five models we integrated existing model training algorithms into our Python library. Each of the models has a set of model hyperparameters that can be adjusted to fit the input data. In Table III we list the parameter values for each model and dataset used in our experimental evaluation.

Feature extraction parameters. In our Python library, we include four FeatureSet objects that implement the feature extraction strategy of a MIA adversary. The F_{Hist} , F_{Corr} , and F_{Ens} feature sets include a pre-processing step in which continuous columns are binned. The number of bins is a configurable parameter that can be adjusted to fit the input data. In our experiments, we set the number of bins to 45, 25, and 25 for the Adult, Texas, and German dataset, respectively.

D. High variance in privacy gain across target records

Fig. 10 shows that the expected gain of 50 randomly chosen target records varies substantially depending on the generative model and feature set targeted by the attack. CTGAN models yield the highest variance from 0.2 to 0.28 under an attack based on the F_{Naive} feature set. Furthermore, as an analysis of the intersection between target records with the lowest expected gain across models and feature set in Fig. 11 shows, which records remain vulnerable differs substantially depending on the combination of data generating model and attack feature set. The two statistical models that are expected to largely preserve the same features (marginal frequency counts in BayNet and IndHist) have a high overlap in the most vulnerable target records under a fixed attack feature set (Fig. 11 left). The same target groups are likely to be vulnerable if the attacker uses F_{Corr} or F_{Ens} on data produced by BayNet-trained models which are expected to preserve correlation patterns. The overlap between attacks using similar feature sets produced on data by CTGAN is much more variable. This is consistent with the results shown in Fig. 10 and Fig. 7: it is hard to predict which targets will remain vulnerable when the features learned by a model from its training data are unknown.

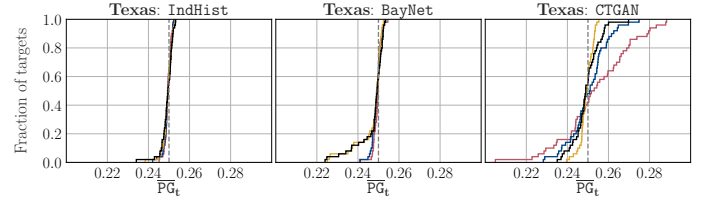


Fig. 10: Cumulative distribution of expected record privacy gain across 50 randomly chosen target records under a Random Forests attack on varying feature sets. Feature sets: F_{Naive} (red), F_{Hist} (blue), F_{Corr} (yellow), F_{Ens} (black).

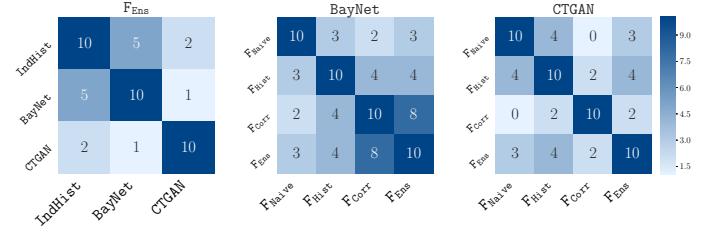


Fig. 11: Pairwise intersection size of the 10 targets with lowest expected privacy gain between different generative models under a fixed attack strategy (left) and between the attack's feature set under a fixed generative model (middle and right). Results for the Texas dataset under a RandForest-based attack.

E. Impact of differentially private network training on changes in data representation

In Fig. 12 we show the impact a worst-case target record's presence has on the structure of the network learned under the BayNet and PrivBay training algorithm, respectively. For each, BayNet and PrivBay, we train two different target models *with a fixed seed* to control changes caused by the random initialisation of the GreedyBayes algorithm during training: one model on a fixed training set without the target and an equivalent model including the target record (labelled OUT and IN). Fig. 12 shows that adding the target record to the training data changes the structure of the network learned by the BayNet algorithm even when the first attribute added to the network is fixed. Differentially private training ensures that the network structure remains the same even when the target is added to the data.

However, Fig. 13 demonstrates that even though the PrivBay algorithm guarantees that the lower-dimensional data representation of the model is not affected by the target's presence, there are visible differences between the average correlations of synthetic data drawn from the two models trained with and without the target, respectively.

TABLE III: Generative Model Hyperparameters

	IndHist nbins	BayNet		CTGAN					
		nbins	degree	embedding_dims	gen_dim	dis_dim	l2scale	batch_size	epochs
Adult	45	45	1	128	(256, 256)	(256, 256)	10^{-6}	500	100
Texas	25	25	1	128	(256, 256)	(256, 256)	10^{-6}	500	100
German	25	25	1	128	(256, 256)	(256, 256)	10^{-6}	100	100

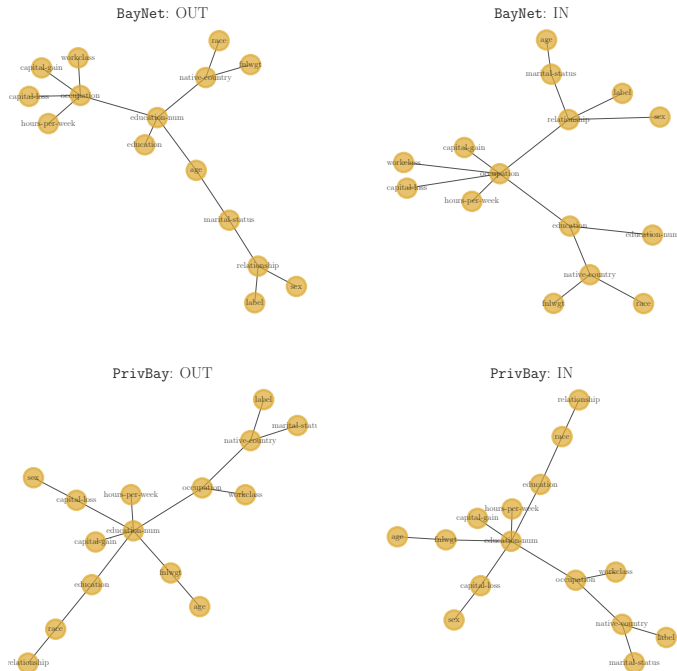


Fig. 12: Bayesian network learned from a fixed training set sampled from the `Adult` dataset with (IN) and without (OUT) a worst-case target record under the two training algorithms `BayNet` and `PrivBay`.

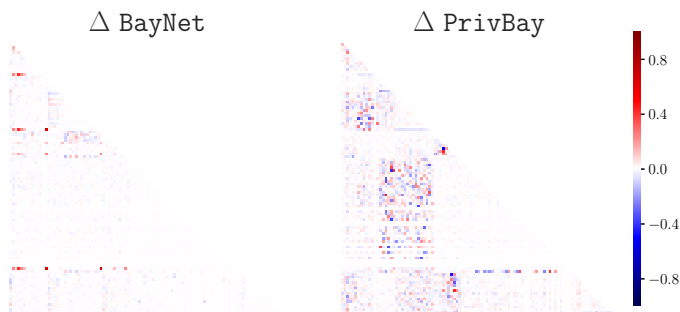


Fig. 13: Difference between the average correlations of 25 synthetic datasets sampled from a fixed target model trained on data including the target and an equivalent model trained without the target. For instance, ΔBayNet shows the delta between synthetic data drawn from the two models labelled BayNet : *OUT* and BayNet : *IN* in Fig. 12. A value of zero indicates that, on average, the target’s presence does not affect how much two attributes are correlated in the synthetic data.